

Physics Unit 7, Part 7B

Irregular cheez ki density (displacement method) aur Matter ki teen halatein · kitab p172 se p175

Seemab, hissa chhota hai magar numericals yahin se bantay hain. Pehle notes mein liquids aur regular solids ka method reh gaya tha (kitab p170, p171), woh Part 7A mein aa raha hai; yahan p172 se p175 tak sab kuch hai.

1. Displacement method (7.2.2, kitab p172, p173)

Kitab ki definition: exam mein bilkul yehi likhna

"We can find the density of **irregular shaped solid objects** like stones which can **sink in the water**. We need **graduated measuring cylinder** to measure volume and **balance** to measure mass of object." Aur aakhir mein: "...any **insoluble** irregular shaped object."

Kitab ke 6 steps (yehi likhne hain, isi tarteeb mein):

1. Find the mass of the irregular shaped stone (balance se).
2. Add some water in measuring cylinder and measure its **initial volume (V_i)**.
3. Tie thread with the object and lower it in the measuring cylinder.
4. Water will rise (displace); measure **final volume (V_f)**. Ye reading paani + pathar dono ka volume hai.
5. Volume of object = $V_f - V_i$
6. Density nikalne ke liye mass ko volume se divide karo.

$$\text{Density of Solid} = \text{Mass of Solid} / \text{Volume of Solid} \quad (\text{Equation 7.4})$$

Figure 7.3 banana aa jaye: do cylinders. Pehle mein sirf paani, level V_i . Doosre mein pathar **dhaage se latka hua** andar, ooncha level V_f .

Example 7.4 (kitab p173): pathar ki density

Mass = 80.52 g, paani 18 mL se 46 mL par aaya.

$$V = V_f - V_i = 46 - 18 = \mathbf{28 \text{ mL} = 28 \text{ cm}^3}$$

$$\rho = 80.52 \text{ g} \div 28 \text{ cm}^3 = \mathbf{2.88 \text{ g/cm}^3}$$

Kitab ki galti: yaad rakho

Kitab ka apna data 28 cm^3 deta hai, magar neeche $80.52 \div 25 \text{ cm}^3$ likh kar jawab 3.22 g/cm^3 chhapa hai. Ye **ghalat** hai. Tum hamesha **28** se divide karna, jawab 2.88 g/cm^3 .

Example 7.3 (kitab p172): eent (brick) ki density, cm se metre

Mass = 3.30 kg; $21.6 \text{ cm} \times 10.2 \text{ cm} \times 6.35 \text{ cm}$.

Pehle cm ko metre: 0.216 m, 0.102 m, 0.0635 m.

$$V = 0.216 \times 0.102 \times 0.0635 = \mathbf{0.00140 \text{ m}^3}$$

$$\rho = 3.3 \text{ kg} \div 0.00140 \text{ m}^3 = 2357.14 = \mathbf{2360 \text{ kg/m}^3}$$

Do pakki baatein

1 mL = 1 cm³, is liye cylinder ki reading seedha cm³ ban jati hai.

Units milao: mass kg mein ho to volume m³ mein (jawab kg/m³); mass g mein ho to volume cm³ mein (jawab g/cm³). kg ko cm³ se kabhi divide mat karna.



2. Matter ki teen halatein (7.3, kitab p174, p175)



Kitab ki definition: exam mein bilkul yehi likhna

"Matter exists in three fundamental states; **solid, liquid and gas**. Matter consists of particles (atoms and molecules). **Gaps between these particles is the main reason that divides the matter in three states**. Particles of matter apply an **attractive force** on each other. The difference in strength of this force is the reason for smaller or large gaps between particles."

Asli farq sirf do cheezein: **force kitni strong** aur **gap kitna**. Shape, density, motion sab usi se nikalte hain.

Cheez	Solid	Liquid	Gas
Attractive force	Strong	Intermediate (solid se kam)	Negligibly weak
Arrangement	Closely packed, fixed lattice pattern	Randomly arranged, phir bhi paas paas	Randomly arranged, faasla particle ke size se bhi bara
Movement	Sirf mean position par vibrate	Ek doosre par flow aur glide , position badalti hai	Tez random motion , aapas mein aur deewar se takraata hai
Shape	Fix shape	No fix shape; container ki shape le leta hai	No fix shape, no fix volume; poora container bhar deti hai
Density	High	Medium (intermediate)	Low
Energy	Low energy	Greater energy	Highest energy

Ye **Table 7.2** + 7.3.1 ke points ek hi jagah. **FYI (p175)**: liquid ke particles free move karte hain magar force ki wajah se container ke **neeche** rehte hain; gas ke phail kar poora container bhar dete hain.



Kitab ki Figure 7.4 ke label ulat gaye hain

"Solids" wale cylinder mein particles **door door phaile** hain (ye gas hai), "Gases" wale mein **ghichi packed** (ye solid hai). **Jahan particles sab se paas paas hain woh solid hai**.



Khud ko check karo (5 sawal, ye paper mein bhi aa sakte hain)

1. Displacement method kis tarah ki cheez ke liye hai, aur us mein kaunsi **do** sharten honi chahiyen?
2. Ek pathar ka mass 175 g hai. Measuring cylinder mein paani 30 mL se 55 mL par aa gaya. Is ka volume aur density nikalo.
3. Solid ki density gas se kahin zyada kyun hoti hai? Attractive force ke lafz mein samjhaao.
4. Kitab ke Example 7.4 mein chhapa jawab 3.22 g/cm^3 hai. Sahi jawab kya hai, aur kitab se galti kahan hui?
5. Table 7.2 mein liquid aur gas dono "randomly arranged" hain. Phir in mein farq kis cheez se karo ge? Do farq likho.



Jawab

1. Irregular shaped solid object ki. Sharten: woh paani mein **sink** kare (doobe) aur **insoluble** ho (paani mein ghule nahin). 2. $V = 55 - 30 = 25 \text{ mL} = 25 \text{ cm}^3$; $\rho = 175 \div 25 = \mathbf{7.0 \text{ g/cm}^3}$. 3. Solid mein attractive force **strong** hai, is liye particles closely packed aur fixed lattice mein hain, gaps bohat chhote. Utne hi volume mein zyada mass aa jata hai, to density high. Gas mein force negligible hai, gaps bohat bare, is liye density low. 4. Sahi jawab **2.88 g/cm³**. Kitab ne volume $46 - 18 = 28 \text{ cm}^3$ nikala, magar divide 25 cm^3 se kar diya. 5. (i) **Movement**: liquid ke particles ek doosre par glide karte hain, gas ke har taraf tezi se random motion mein hain. (ii) **Density aur energy**: liquid ki density medium aur energy greater, gas ki density low aur energy highest. Liquid ka volume fix hai, gas ka nahin.